CHOW'S LEMMA WITH AN IMMERSION INTO PROJECTIVE SPACE FOR SEPARATED FINITE TYPE MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9446-9477, immediately following the statement at 9431-9444.

Editorial note. Two 'Clearly' transfer steps (source lines 9450-9452 and 9471-9473) are retained verbatim; the justification recorded in proof_roadmap (composition preserves proper surjective, base change preserves immersions) is the reviewer's checklist, not author text.

Editorial note. All five cross-chapter pointers are Limits of Spaces lemmas recorded in assembly.external_references; the terminal same-chapter pointer is a line-indexed stub into the pinned source.

Allowed standard prerequisites (editorial).

- Finite presentation replacement by a closed immersion. Limits of Spaces, Proposition spaces-limits-proposition-separated-closed-in-finite-presentation, cited at proof lines 9448-9450. X embeds as a closed subspace of X_1 separated and of finite presentation over Y; proper surjective X' -> X composed with the closed immersion transfers the conclusion back.
- Approximation of separated quasi-compact spaces by finite type layers. Limits of Spaces, Proposition spaces-limits-proposition-approximate, cited at proof lines 9455-9457. Writes Y = lim Y_i with Y_i quasi-separated of finite type over Spec(Z), the ladder on which everything descends.
- Descending separatedness and finite presentation along the limit. Limits of Spaces, Lemmas spaces-limits-lemma-descend-separated (9457-9459), spaces-limits-lemma-descend-finite-presentation (9462-9465), spaces-limits-lemma-descend-separated-morphism (9468-9470). Produces an index i and X_i -> Y_i of finite presentation, separated, with X = Y x_{Y_i} X_i.

Source reference. Original lemma 088U: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9275–9275. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SK: focused-seeds/geometry/S9-spaces-chow-finite-type-separated/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084T: focused-seeds/geometry/S9-spaces-chow-finite-type-separated/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0851: focused-seeds/geometry/S9-spaces-chow-finite-type-separated/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SU: focused-seeds/geometry/S9-spaces-chow-finite-type-separated/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0873: focused-seeds/geometry/S9-spaces-chow-finite-type-separated/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f separated of finite type, and Y separated and quasi-compact. Then there exists a commutative diagram*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longrightarrow & \mathbf{P}^n_Y \\ & \searrow & \downarrow & \swarrow & \\ & & Y & & \end{array}$$

where $X' \to X$ is proper surjective morphism and the morphism $X' \to \mathbf{P}^n_Y$ is an immersion.

Proof. By Limits of Spaces, Proposition 0873 we can find a closed immersion $X \to X_1$ where X_1 is separated and of finite presentation over Y. Clearly, if we prove the assertion for $X_1 \to Y$, then the result follows for X. Hence we may assume that X is of finite presentation over Y.

We may and do replace the base scheme S by $\mathrm{Spec}(\mathbf{Z})$. Write $Y = \lim_i Y_i$ as a directed limit of quasi-separated algebraic spaces of finite type over $\mathrm{Spec}(\mathbf{Z})$, see Limits of Spaces, Proposition 07SU. By Limits of Spaces, Lemma 084T we may assume that Y_i is separated for all i. By Limits of Spaces, Lemma 07SK we can find an index $i \in I$ and a scheme $X_i \to Y_i$ of finite presentation so that $X = Y \times_{Y_i} X_i$. By Limits of Spaces, Lemma 0851 we may assume that $X_i \to Y_i$ is separated. Clearly, if we prove the assertion for X_i over Y_i, then the assertion holds for X. The case $X_i \to Y_i$ is treated by Lemma 088U. □